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ELEC 366 Communication LAB
Experiement 2 Prestudy Questions

1) Angle modulation is a technique use in communication systems to encode information (message signal) on to a carrier wave by varying the phase or frequency of the carrier signal. This modulation techique is used in radio braadcasting, radar systems, space communication, etc...
There are two main types of angle modulation:

1. Frequency modulation

In FM, the frequency of the carrier wave is varied in proportion to the amplitude of the modulating signal. Assume m(t) is the message signal.
$y_{FM}(t) = A_c \cos(2\pi f_c t + \phi(t))$ and $\phi(t) = 2\pi k_f \int_{-\infty}^{t} m(z)dz$

so $y_{FM}(t) = A_c \cos(2\pi f_c t + 2\pi k_f \int_{-\infty}^{t} m(z)dz)$

$f_i(t) = f_c + k_f m(t)$ $\Delta f = k_f m_p$ $m_p = \max|m(t)|$

2. Phase modulation

In PM, the phase of the carrier wave is varied in propotion to the amplitude of the modudating signal.
$y_{PM}(t) = A_c \cos(2\pi f_c + \phi(t))$ and $\phi(t) = k_p m(t)$
so $y_{PM}(t) = A_c \cos(2\pi f_c + k_p m(t))$ and $f_i(t) = f_c + \frac{k_p}{2\pi} \dot{m}(t)$

- PM is less commonly ased compared to FM but finds applications in certain communication systems and radar systems where phase information is crucial.
- It is less sensitive to frequency variations but requires more bandwith compared to FM.

2) Main difference in angle modulation, information is encoded by varying the phase or frequency of the carrier wave when in Amplitude modulation information is encodeol by varying the amplitude of the carrier wave

Advantages of Angle modulation over Amplitude modulation.
- Noise Immunity: Angle modulation, offer better noise immunity because they have a constant amplitude
- Reduced Interference since the angle modulation techniques do not rely on variations in amplitude

Disavantages:
- Bandwidth Requirements: FM and PM requires wider bandwith to transmit signals compared to AM.
- Complexity and Cost: Implementing angle modulation techniques especially FM, requries more complex circuitry and modulation/demodulation shemes compared to AM. So it make it higher cost related to AM.
- Synchronization issues:
PM modulation in particular, can be sensitive to synchronization issues, espesially in systems where precise phase info is crucial

3) $f_m = 2\,kHz$ $\quad \Delta f = 5\,kHz$

a) $y(t) = A_c \cos(2\pi f_c t + \phi(t))$

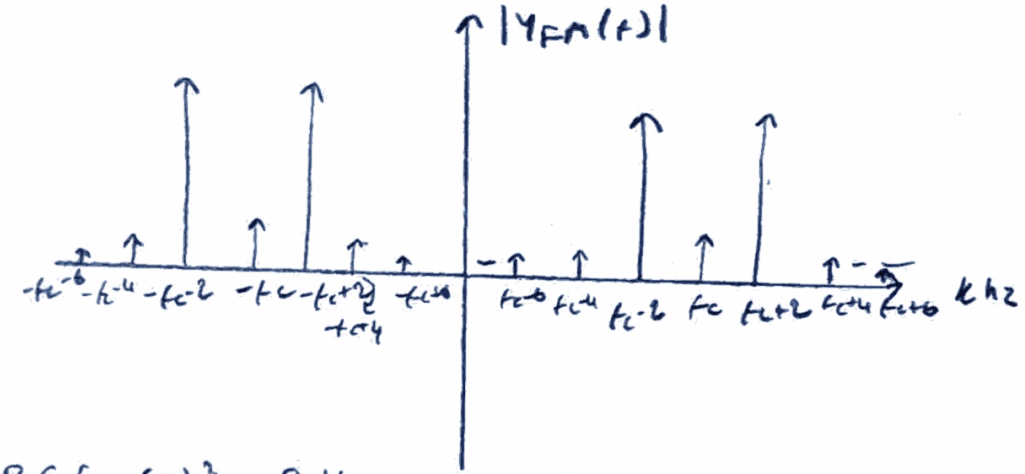
FM: $\phi(t) = 2\pi k_f \int_{-\infty}^{t} m(z)\,dz$

$\Rightarrow y(t) = A_c \cos(2\pi f_c t + 2\pi k_f \int_{-\infty}^{t} m(z)\,dz)$

$m(t) = a\cos(2\pi f_m t)$ olsun

$y_{FM}(t) = A_c \cos(2\pi f_c t + \beta \sin(2\pi f_m t))$, $\beta = \frac{k_f \cdot a}{f_m} = \frac{\Delta f}{B} = \frac{5}{2}$

b) $= A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos(2\pi(f_c + n f_m)t)$



c) $BG\{m(t)\} = B\,Hz = f_m = 2\,kHz$

Carson rule: $BG = 2B(\beta+1)$ when $\beta = \frac{\Delta f}{B} = \frac{5\,kHz}{2\,kHz} = \frac{5}{2}$

$BG = 2 \cdot 2k\left(\frac{7}{2}\right)$

$= 14\,kHz$ //

$\beta = \frac{k_f \cdot a}{f_m}$ for sinusoidal signals

d) If the amplitude of message signal is increased three times and f_m decreased by half

$\beta_{new} = \frac{k_f \cdot 3a}{0,5 f_m} = \frac{6 k_f a}{f_m} = 6\beta = 15$ //

$\beta_{new} = \frac{\Delta f_{new}}{B_{new}}$ $\qquad 15 = \frac{\Delta f_{new}}{0,5 f_m}$ $\Rightarrow \Delta f_{new} = 15\,kHz$

$BG_{new} = 2 B_{new}(\beta_{new}+1) = 2 \cdot 1\,(15+1) = 32\,kHz$ //